

1 Performance profile

Overview

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Transaction code	effpf
Function authorization	effpf

The *Performance profile* application provides a tabular and graphic presentation of the production performance interrelationships. The collected performance data is compressed to days and shifts. This way, the application can provide indispensable production KPIs for all persons in a position of responsibility. This application provides a clear overview of all quantities and durations that are necessary to reliably assess the production status.

Selection criteria

The application provides the following selection criteria:

Workplace

This selection criterion refers to the workplace in the machine or workplace master data. You can also use wildcards (placeholders *).

Group

This selection criterion refers to the group in the machine or workplace master data. The application shows all machines or workplaces that are assigned to the selected group. You can also use wildcards.

Date from ... to ...

Fill in the fields from/to to restrict the period to be evaluated

Shift / time

Restrict the defined period (date from/to). Select shifts or specify a time (from - until).

Report group

This selection criterion refers to the report groups. The application shows all workplaces/machines assigned to the selected report group.

Company

This selection criterion refers to the company defined in the machine or workplace master data. The application shows all machines or workplaces that are assigned to the selected company. You can also use wildcards.

Responsibility area

This selection criterion refers to the responsibility area in the machine master data. Note: The user can only view those machines that are included in the responsibility areas that are assigned to the user.

Short name

This selection criterion refers to the short designation of machines in the master data. The application shows all machines or workplaces matching the entered string. You can also use wildcards.

Cost center

This selection criterion refers to the cost center stored in the machine or in the workplace master data. All machines or workplaces are displayed that are assigned to the selected cost center. You can also use wildcards.

Name (designation)

This field refers to the name of machines and workplaces defined in the machine master data. The application only shows those machines matching the entered string. You can also use wildcards (placeholders *).

Selection options

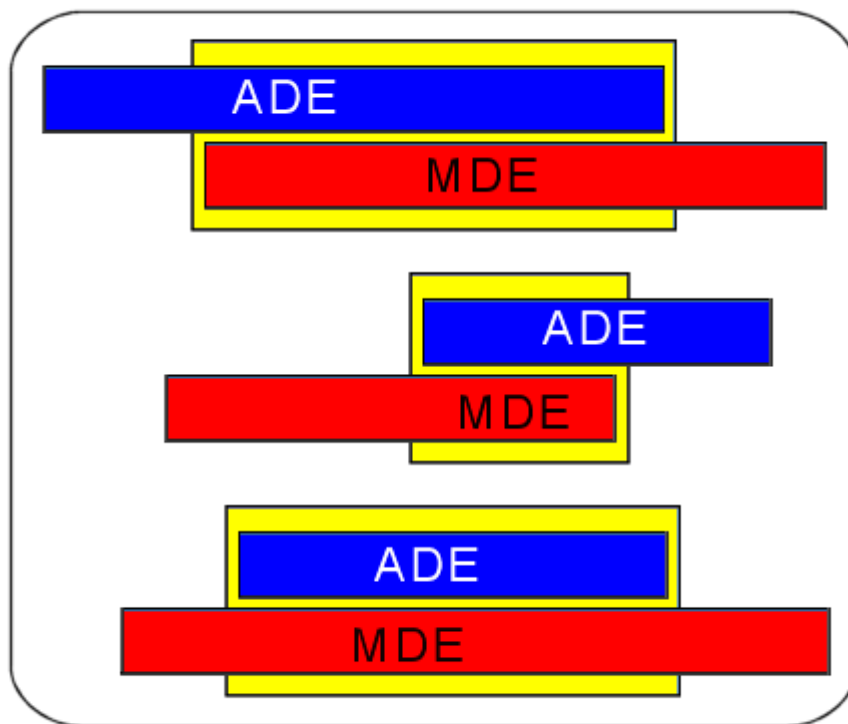
- No selection:* If you enable this option, you cannot select the other selection criteria (order and resource).
- Order:* You can use order parameters to restrict the data.
- Resource:* You can use resource parameters to restrict the data.

Order/final article/batch number

If you select these options, the application only includes completed ADE postings. If the order is currently still running on the machine, the system does not take into account the time period between the last logon and now. Therefore it is possible that differences appear between the machine evaluation and the order-related evaluation. The application only includes ADE postings that coincide with the evaluation period. If necessary, choose a selection period making sure that the required ADE postings coincide with this period. For this order-related evaluation, MPDV recommends selecting data via the "shift" option instead of the "time".

The order number, article/item or the batch (from the order header) may be used as selection criteria.

The illustration below shows an example of how ADE and MDE postings can overlap. The ADE postings take priority in this evaluation type. The yellow areas show the result of this evaluation. MDE quantities and durations are calculated on a pro rata basis to achieve the result.



If several orders are logged on to the machine simultaneously, this evaluation type (select one of the options *order*, *final article* or *batch number*) assigns the full machine time (yellow area) and quantities to every order. The fact that orders are run in parallel will not result in a proportionate calculation.

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If the selection period exceeds the period for the online data area, the system applies the implicit solution and selects the medium-term data area as well. Therefore, there is no need for an explicit activation in order to be able to access the medium-term data set.

Resource/resource type

Restrict the selected posting records integrated in the evaluation by selecting the logged in resources or resources of a specific resource type.

Group result

Use this selection criterion to group the results according to the following parameters:

- Year
- Month
- Calendar week
- Date
- Shift date and shift

Performance profile detail application

The performance profile of machine-related performance data is presented for a specific period of time and a certain number of workplaces. The result depends on the selection made and therefore on the selection criteria available in the selection panel.

Date category

Depending on the selection of the option "Group results" in the selection panel, the columns are completed as shown below (the columns that are not listed for the respective selection remain empty):

- Select shift date and shift: the application shows the year, calendar week number, month, shift date, shift.
- Select date: the application shows the year, calendar week number, month, shift date.
- Select week: the application shows the year, calendar week number.
- Select month: the application shows the year, month.
- Select year: the applications shows the year.

Primary quantity, secondary quantity, tertiary quantity, basic quantity category

Workplace/ machine-related quantities recorded in the corresponding quantity types

- Yield
- Scrap
- Rework
- Open quantity

or the relevant quantity units (if relevant in the customer system).

Durations category

- Production = RPA11
- Downtime = RPA01 + RPA02 + RPA03 + RPA04 + RPA05 + RPA06 + RPA07 + RPA08 + RPA09 + RPA10
- Total = production + downtime

Key figures category

- Rate of capacity utilization

The rate of capacity utilization is the quotient deriving from effective run time and machine working time.

Rate of capacity utilization = $100 / (RPA01 + RPA02 + RPA03 + RPA04 + RPA05 + RPA06 + RPA07 + RPA08 + RPA09 + RPA10 + RPA11) * RPA11$

Formula definition: $rpa11 / (rpa1+rpa2+rpa3+rpa4+rpa5+rpa6+rpa7+rpa8+rpa9+rpa10+rpa11) * 100$

Use the formula **rcu** to customize the calculation. If the formula management does not include this formula, then you have to create the formula in order to change the calculation.

- Assignment utilization rate:

The assignment utilization rate represents the relationship deriving from the effective run time (main utilization time) and the adjusted machine operation time (i.e. without planned downtimes).

Assignment utilization rate = $100 / (\text{total of RPA 1 to RPA 11 minus RPA 6}) * RPA 11$

Formula definition: $rpa11 / (rpa1+rpa2+rpa3+rpa4+rpa5+rpa7+rpa8+rpa9+rpa10+rpa11) * 100$

Use the formula **ocu** to customize the calculation. If the formula management does not include this formula, then you have to create the formula in order to change the calculation.

- Efficiency (efficiency is only available when combined with the "lines" license)

Efficiency is the quotient derived from the effective run time and the general run time, meaning the sum total of the effective run time and any interruptions as a result of machine-related disturbances or unscheduled shutdowns. Any other disturbances, e.g. organization-related disturbances, are not taken into account here.

Efficiency = $100 / (RPA02 + RPA05 + RPA11) * RPA11$

- Techn. efficiency

The reference value for the technical efficiency is the sum total of the effective run time and interruptions because of technical (machine-related) disturbances. Other interruptions (e.g. of a organizational nature) are not considered: Techn. efficiency = $100 / (RPA02 + RPA11) * RPA11$

Formula definition: $rpa11 / (rpa2 + rpa11) * 100$

Use the formula **tec_ef** to customize the calculation. If the formula management does not include this formula, then you have to create the formula in order to change the calculation.

- Rate = $100 / ((\text{yield (primary quantity)} + \text{scrap (primary quantity)} + \text{rework quantity (primary quantity)} + \text{open quantity (primary quantity)}) * \text{yield (primary quantity)})$.

Formula definition: $\text{yield.primary} / (\text{yield.primary} + \text{scrap.primary} + \text{rework.primary} + \text{problem.primary}) * 100$

Use the formula **yie_ra** to customize the calculation. If the formula management does not include this formula, then you have to create the formula in order to change the calculation.

- Scrap rate = $100 / ((\text{yield (primary quantity)} + \text{scrap (primary quantity)}) * \text{scrap (primary quantity)})$

Formula definition: $\text{scrap.primary} / (\text{yield.primary} + \text{scrap.primary} + \text{rework.primary} + \text{problem.primary}) * 100$

Use the formula **scr_ra** to customize the calculation. If the formula management does not include this formula, then you have to create the formula in order to change the calculation.

Quantities detail application

This chart shows, based on the selected entries in the table, the relationship between the quantities.

Depending on the option "group result", that you have enabled in the selection range, the values are displayed per

- Shift/shift date (select shift date and shift)

	1	2	3
	10.01.2011	10.01.2011	10.01.2011
- Shift date (select shift date)

09.01.2011	10.01.2011	11.01.2011
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- Calendar week (select calendar week)

2011 Woche: 1	2011 Woche: 2	2011 Woche: 3
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- Month (select month)

2011/1	2011/2
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- Year (select year)

2010	2011
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You can use the "Displayed series" combo box to select which quantities you would like to have displayed (e.g. only yield and scrap).

Durations detail application

The durations detail application shows, based on the selected entries in the table, the relationship between the production duration and downtime. Depending on the option "group result", that you have enabled in the selection range, the values are displayed per

- Shift/shift date (select shift date and shift)

- Shift date (select shift date)
- Calendar week (select calendar week)
- Month (select month)
- Year (select year)



This detail application is docked right behind the key figures detail application.

Key figures detail application

The key figures detail application shows, based on the selected entries in the table, the efficiency or the technical efficiency, assignment utilization rate, rate, scrap rate or the rate of capacity utilization in graphic form.

Select the relevant key figure from the combo box. If no key figure is selected, the rate of capacity utilization is shown.

Depending on the option "group result", that you have enabled in the selection range, the key figures are displayed per

- Shift/shift date (select shift date and shift)
- Shift date (select shift date)
- Calendar week (select calendar week)
- Month (select month)
- Year (select year)

The key figure in the label (if activated) is displayed to the second decimal place.



This detail application is docked right behind the durations detail application.